

# Energy-Based Reconstruction of a Protein Backbone from its $\alpha$ -Carbon Trace by a Monte-Carlo Method

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**Abstract:** An automatic procedure is proposed for reconstruction of a protein backbone from its  $C^\alpha$ -trace; it is based on optimization of a simplified energy function of a peptide backbone, given its  $\alpha$ -carbon trace. The energy is expressed as a sum of the energies of interaction between backbone peptide groups that are not neighbors in the sequence, the energies of local interactions within all amino acid residues, and a harmonic penalty function accounting for the conservation of standard bond angles. The energy of peptide group interactions is calculated using the assumption that each peptide group acts as a point dipole. For local interaction energy, use is made of a two-dimensional Fourier series expansion of the energies of model terminally blocked amino acid residues, calculated with the Empirical Conformational Energy Program for Peptides (ECEPP/3) force field in the angles  $\lambda^{(1)}$  and  $\lambda^{(2)}$  defining the rotation of peptide groups adjacent to a  $C^\alpha$  carbon atom about the corresponding  $C^\alpha \dots C^\alpha$  virtual-bond axes. To explore all possible rotations of peptide groups within a fixed  $C^\alpha$ -trace, a Monte Carlo search is carried out. The initial  $\lambda$  angles are calculated by aligning the dipoles of the peptide groups that are close in space, subject to the condition of favorable local interactions. After the Monte Carlo search is accomplished with the simplified energy function, the energy of the structure is minimized with the ECEPP/3 force field, with imposition of distance constraints corresponding to the initial  $C^\alpha$ -trace geometry. The procedure was tested on model  $\alpha$ -helices and  $\beta$ -sheets, as well as on the crystal structure of the immunoglobulin binding protein (PDB code: 1IGD, an  $\alpha/\beta$  protein). In all cases, *complete* backbone geometry was reconstructed with a root-mean-square (rms) deviation of 0.5 Å from the all-atom target structure.

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**Key words:** Monte Carlo method; protein folding; conversion to all-atom backbone; dipole alignment; hydrogen bonding

## Introduction

Reconstruction of an all-atom polypeptide chain from its coarse-grain representations in which only the positions of its  $\alpha$ -carbon atoms and, sometimes in addition to this, the average positions of side-chain centers are known has been a subject of extensive studies for many years,<sup>1–9</sup> because many experimental and theoretical approaches, such as low-resolution crystallography, or prediction of protein structure by threading or *ab initio* methods result in low-resolution protein structures expressed in terms of  $\alpha$ -carbon and, additionally, side-chain centroid coordinates rather than in structures at atomic resolution. Especially in the case of an *ab initio* prediction of protein structure it is impossible to explore the conformational space at the all-atom level, and it is essential to use a reduced representation.<sup>10</sup> Because the ultimate goal of prediction is to obtain structures at the all-atom resolution, it is necessary to convert coarse-grain structures into all-atom structures, without violating the  $C^\alpha$ -trace.

There are three types of methods for the reconstruction of all-atom chains from  $\alpha$ -carbon traces: (i) methods that use fragment libraries<sup>2–5</sup> or other information from the Protein Data Bank

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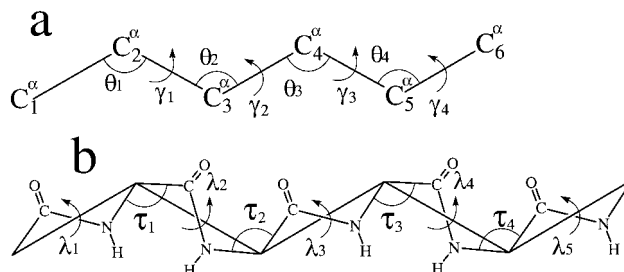
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(PDB)<sup>11</sup>; (ii) methods based on geometric criteria (i.e., positioning the peptide groups in the  $C^\alpha$ -trace to obey the constraints arising from the bond-length and bond-angle geometry and the planarity of the peptide groups);<sup>1,6</sup> and (iii) methods based on optimizing the energy of the backbone,<sup>8,12</sup> given a  $C^\alpha$  trace. Methods of the last class are especially desirable for the conversion of low-resolution structures obtained by using energy-based prediction procedures, such as that which makes use of the united-residue (UNRES) force field,<sup>10,13–21</sup> which is being developed in our laboratory. A few years ago we designed an algorithm of the latter class, hereafter referred to as the *dipole path* method.<sup>8</sup> Its underlying principle is the optimum alignment of the peptide group dipoles, given the condition that the orientations of the peptide groups in the chain must be those in real protein structures; the latter implicitly avoided bad local conformational states of individual amino acid residues. Then the lowest energy configuration (optimal alignment) of the dipoles is sought for each path. The dipole path method performs very well for  $\alpha$ -helical proteins, for which the dipole path network is easy to define; however, it is difficult to apply it to  $\beta$ -sheet proteins, because automatic determination of the directions in which the dipoles should be aligned is not straightforward, and usually manual correction of the network is necessary. Moreover, electrostatic alignment does not determine the positions of peptide groups in loop regions sufficiently well. Thus, the dipole path method is difficult to apply in practice for proteins of other than the  $\alpha$  structural class, and does not determine the orientations of peptide groups in loop regions with sufficient reliability. The latter implies that more computer time is spent on the refinement of the resulting all-atom structures using, for example, the electrostatically driven Monte Carlo method.<sup>22–25</sup>

The purpose of the present work is to design a fully automatic energy-based method for the reconstruction of a polypeptide backbone, thereby surmounting the shortcomings of the dipole path method. Instead of an elaborate construction of a dipole path network, we *explicitly* optimize a simplified energy of the backbone, which includes both dipole-electrostatic and backbone-local terms. The performance of the method is illustrated on test examples, which include a real  $\alpha/\beta$  protein. As in our previous work,<sup>8,13</sup> we assume that, for a given  $C^\alpha$  trace, the problem of reconstruction of an all-atom chain from a united-residue chain can be separated into two separate tasks: (i) reconstruction of the all-atom backbone, and (ii) reconstruction of the side-chain geometry for a given backbone. This is justified by the observation that the backbone geometry of well-defined secondary-structure elements ( $\alpha$ -helices,  $\beta$ -sheets,  $\beta$ -turns, etc.) does not vary significantly with the type of amino acid residue. In our current treatment, we determine the side-chain geometry by varying the consecutive dihedral angles  $\chi$  of all consecutive side chains and searching for the lowest energy of the system as a function of the dihedral angle currently scanned.<sup>15</sup> This, however, does not conserve the positions of side-chain centroids in the parent united-residue structure, which would be desirable. This issue will be dealt with in our further work.



**Figure 1.** (a) Virtual geometry of a polypeptide chain of  $n$   $C^\alpha$ -atoms. The geometry is defined by virtual-bond valence angles  $\theta_1, \theta_2, \dots, \theta_{n-2}$  and virtual-bond dihedral angles  $\gamma_1, \gamma_2, \dots, \gamma_{n-3}$ . (b) Same polypeptide chain after positioning the peptide groups whose orientation is defined by the angles  $\lambda_1, \lambda_2, \dots, \lambda_{n-1}$  for rotation about the corresponding  $C^\alpha \cdots C^\alpha$  virtual bonds. The rotations are restricted to satisfy the real  $N-C-C'$  bond angles  $\tau_1, \tau_2, \dots, \tau_{n-2}$ . Reconstruction of the backbone means finding the angles  $\lambda_1, \lambda_2, \dots, \lambda_{n-1}$  that satisfy these conditions and result in a low intrabackbone interaction energy.

## Methods

### Formulation of the Problem

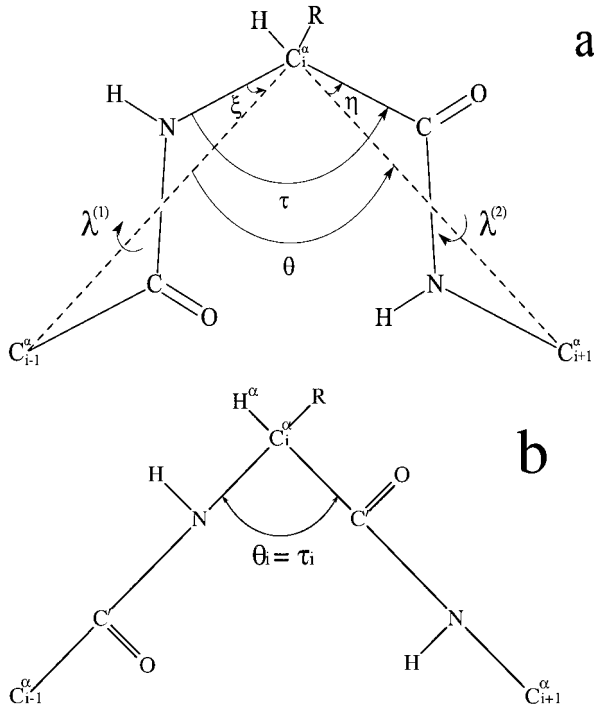
Following the UNRES model of polypeptide chains,<sup>10,13–21</sup> the geometry of a  $C^\alpha$ -trace consisting of  $n$   $C^\alpha$  atoms is defined in terms of the virtual-bond valence angles  $\theta_1, \theta_2, \dots, \theta_{n-2}$  and virtual-bond dihedral angles  $\gamma_1, \gamma_2, \dots, \gamma_{n-3}$ , describing the rotation about the virtual bonds between consecutive  $C^\alpha$  atoms. All virtual  $C^\alpha \cdots C^\alpha$  bonds are assumed to have a fixed length of 3.8 Å, corresponding to *trans* polypeptide groups (Fig. 1). As pointed out in the Introduction, in this work we do not consider side chains.

Reconstruction of an all-atom backbone from a given  $C^\alpha$  trace means finding the positions of the peptide groups between the consecutive  $\alpha$ -carbon atoms allowed by the valence bond-angle constraints. These positions are most conveniently defined in terms of the angles  $\lambda$  for rotation of peptide groups about the respective  $C^\alpha \cdots C^\alpha$  virtual-bond axes, introduced by Nishikawa et al.<sup>26</sup> (Fig. 2). For a chain segment composed of three consecutive  $\alpha$ -carbon atoms centered at  $C_i^\alpha$  we define two angles  $\lambda_i^{(1)}$  and  $\lambda_i^{(2)}$  as follows.  $\lambda_i^{(1)}$  denotes the angle of clockwise rotation (when looking from  $C_i^\alpha$  toward  $C_{i-1}^\alpha$ ) of the peptide group  $p_{i-1}$  (lying between  $C_{i-1}^\alpha$  and  $C_i^\alpha$ ), with  $\lambda_i^{(1)} = 0$  when  $C_{i+1}^\alpha$  belongs to the plane of the peptide group  $p_{i-1}$  and lies on the same side of the  $C_{i-1}^\alpha \cdots C_i^\alpha$  axis as the oxygen of this peptide group. Similarly,  $\lambda_i^{(2)}$  is the angle of clockwise rotation (when looking from  $C_i^\alpha$  toward  $C_{i+1}^\alpha$ ) of the peptide group  $p_i$ , with  $\lambda_i^{(2)} = 0$  when  $C_{i-1}^\alpha$  lies in the plane of peptide groups  $p_i$ , on the opposite side of the  $C_i^\alpha \cdots C_{i+1}^\alpha$  axis from the oxygen atom of this peptide group. As can easily be realized, not all combinations of angles  $\lambda^{(1)}$  and  $\lambda^{(2)}$  are allowed, because of the constraint imposed by a fixed value of the valence-bond angle  $\tau$  centered at the middle  $C^\alpha$  atom.<sup>1,8,26</sup>

Now, for each peptide group  $p_i$ ,  $i = 1, 2, \dots, n-1$ , we define a unique angle  $\lambda$  as follows:

$$\lambda_i = \lambda_{i+1}^{(1)}, i = 1, 2, \dots, n-2$$

$$\lambda_{n-1} = \lambda_{n-1}^{(2)} \quad (1)$$



**Figure 2.** (a) Definition of the dihedral angles  $\lambda^{(1)}$  and  $\lambda^{(2)}$  for rotation of the peptide groups about the  $C^\alpha \cdots C^\alpha$  virtual bonds of a peptide unit<sup>26</sup> and of the planar angles  $\xi$ ,  $\eta$ , and  $\tau$ . (b) A linearized peptide group with  $\xi = \eta = 0$  and  $\tau = \theta$ .

The angles  $\lambda_1, \lambda_2, \dots, \lambda_{n-1}$  will hereafter be used as variables defining the orientation of peptide groups (Fig. 1b).

### Energy Function

We seek the values of the angles  $\lambda_1, \lambda_2, \dots, \lambda_{n-1}$  that minimize the energy of the backbone. This energy can be expressed as a sum of the energy of electrostatic and van der Waals interactions ( $E_{el}$  and  $E_{vdw}$ , respectively) between the atoms of the peptide groups that are not neighbors in the chain, a local-interaction term ( $E_{loc}$ ), which includes all noncovalent interactions within peptide residues, and a bending term accounting for the distortion of the valence-bond angles  $\tau$  ( $E_{bend}$ ; Fig. 1b), as expressed by eq. (2). The term  $E_{bend}$  is introduced to keep the angles  $\tau$  at the standard values used in the ECEPP/3 force field,<sup>27</sup> which is implemented in the subsequent stage of refinement (unrestricted variation of the angles  $\lambda^{(1)}$  and  $\lambda^{(2)}$  would lead to variation of  $\tau$ <sup>26</sup>).

$$E = E_{el} + E_{vdw} + E_{loc} + E_{bend} = \sum_{i=1}^{n-3} \sum_{j=i+2}^{n-1} E_{el;ij} + \sum_{i=1}^{n-3} \sum_{j=i+2}^{n-1} E_{vdw;ij} + \sum_{i=2}^{n-1} E_{loc;i} + \sum_{i=2}^{n-1} E_{bend}(\tau_i) \quad (2)$$

In the computation of  $E_{el}$  and  $E_{vdw}$ , we do not consider neighboring peptide groups because the corresponding energy of

electrostatic interaction is included in the local interaction energy  $E_{loc}$ .

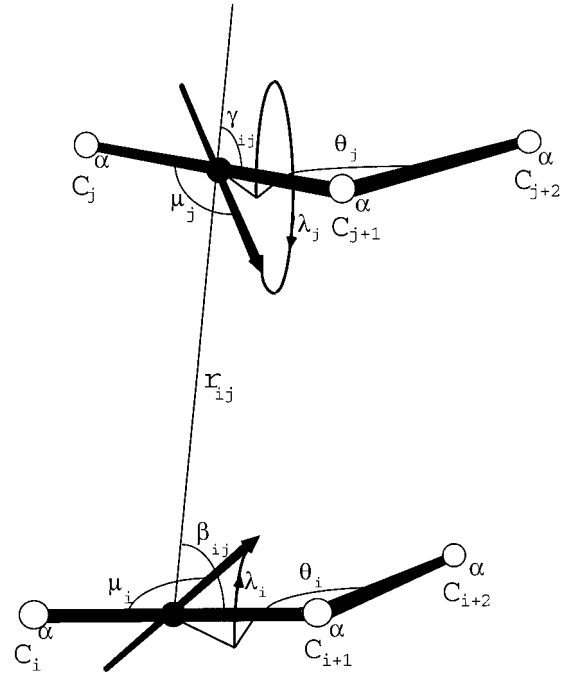
As in our earlier work,<sup>8</sup> to compute the electrostatic component of the energy of peptide-group interaction, we represent each peptide group as a point dipole located in the middle between the consecutive  $C^\alpha$  atoms and lying in the peptide-group plane; the dipole moment is computed according to the charge distribution on the peptide-group atoms<sup>8</sup> (Fig. 3). Then the energy of electrostatic interaction between peptide groups  $p_i$  and  $p_j$  is expressed by eq. (3).<sup>28</sup>

$$E_{el;ij} \approx E_{dip;ij} = \frac{\mathbf{p}_i \cdot \mathbf{p}_j - 3(\mathbf{p}_i \cdot \mathbf{e}_{r_{ij}})(\mathbf{p}_j \cdot \mathbf{e}_{r_{ij}})}{\epsilon r_{ij}^3} = \mathbf{e}_{r_{ij}}^T \mathbf{A}_{ij} \mathbf{e}_{p_j} \quad (3)$$

with

$$\mathbf{A}_{ij} = \frac{p_i p_j}{\epsilon r_{ij}^3} (\mathbf{I} - 3\mathbf{e}_{r_{ij}} \mathbf{e}_{r_{ij}}^T) \mathbf{T}_{ij} \quad (4)$$

where  $\mathbf{T}_{ij}$  is a matrix transforming the reference system of peptide group  $p_j$  to that of  $p_i$ ,  $\mathbf{p}_i$  and  $\mathbf{p}_j$  are the vectors of the dipole



**Figure 3.** Dipole model of interacting peptide groups. The relative orientation of the virtual bonds  $C_i^\alpha \cdots C_{i+1}^\alpha$  and  $C_j^\alpha \cdots C_{j+1}^\alpha$  is described by the angles  $\alpha_{ij}$ ,  $\beta_{ij}$ , and  $\gamma_{ij}$ , defined by eqs. (11), (12), and (13), respectively. The angle  $\alpha_{ij}$  is not shown here because the two virtual bonds  $C_i^\alpha \cdots C_{i+1}^\alpha$  and  $C_j^\alpha \cdots C_{j+1}^\alpha$  are not necessarily coplanar.  $\theta$  is the angle between two successive virtual bonds. The peptide group dipole moments are represented by arrows (pointing from the carbonyl oxygen to the amide hydrogen of a peptide group), and the angles  $\mu_i$  and  $\mu_j$  between them and the virtual bonds are also shown, as well as the rotation angles  $\lambda_i$  and  $\lambda_j$  of the peptide group dipoles. The two dipoles are separated by a distance  $r_{ij}$ .

moments of peptide groups  $p_i$  and  $p_j$ ,  $r_{ij}$  is the distance between the centers of the virtual bonds  $C_i^\alpha \cdots C_{i+1}^\alpha$  and  $C_j^\alpha \cdots C_{j+1}^\alpha$ ,  $\mathbf{e}_{r_{ij}}$  is the unit vector pointing from  $p_i$  to  $p_j$ ,  $\mathbf{e}_{p_i}$  is the unit vector of the dipole moment of the peptide group  $p_i$ , and  $\epsilon$  is the dielectric constant.

For a given configuration of  $C^\alpha$ 's,  $E_{\text{el};ij}$  is a function of the angles  $\lambda_i$  and  $\lambda_j$ . In further discussion, we will consider only those components of the dipole moment vectors that are perpendicular to the  $C^\alpha \cdots C^\alpha$  virtual-bond axes. Then  $E_{\text{dip}}$  can be expressed by eq. (5).<sup>8,16</sup>

$$E_{\text{dip};ij} = e_{ij}(\lambda_i, \lambda_j) = \zeta_{ij} \cos[\lambda_i + \lambda_j - \Lambda_{ij}^{(1)}] - \tilde{\zeta}_{ij} \cos[\lambda_i - \lambda_j - \Lambda_{ij}^{(2)}] \quad (5)$$

with:<sup>8,16</sup>

$$\zeta_{ij} = \frac{\|\mathbf{p}_i\| \|\mathbf{p}_j\| \sin \mu_i \sin \mu_j}{\epsilon r_{ij}^3} Z_{ij}^{(1)} = \frac{B_{ij}^{1/2}}{r_{ij}^3} Z_{ij}^{(1)} \quad (6)$$

$$\tilde{\zeta}_{ij} = \frac{\|\mathbf{p}_i\| \|\mathbf{p}_j\| \sin \mu_i \sin \mu_j}{\epsilon r_{ij}^3} Z_{ij}^{(2)} = \frac{B_{ij}^{1/2}}{r_{ij}^3} Z_{ij}^{(2)} \quad (7)$$

$$Z_{ij}^{(1)} = \frac{1}{2} \sqrt{4(1 + \cos \alpha_{ij}) + W_{ij}^2 - 3(\cos \beta_{ij} + \cos \gamma_{ij})^2} \quad (8)$$

$$Z_{ij}^{(2)} = \frac{1}{2} \sqrt{4(1 - \cos \alpha_{ij}) + W_{ij}^2 - 3(\cos \beta_{ij} - \cos \gamma_{ij})^2} \quad (9)$$

$$W_{ij} = \cos \alpha_{ij} - 3 \cos \beta_{ij} \cos \gamma_{ij} \quad (10)$$

$$\cos \alpha_{ij} = \mathbf{v}_i \cdot \mathbf{v}_j \quad (11)$$

$$\cos \beta_{ij} = \mathbf{v}_i \cdot \mathbf{e}_{r_{ij}} \quad (12)$$

$$\cos \gamma_{ij} = \mathbf{v}_j \cdot \mathbf{e}_{r_{ij}} \quad (13)$$

$$\tan \Lambda_{ij}^{(1)} = \frac{a_{23} + a_{32}}{a_{22} - a_{33}} \quad (14)$$

$$\tan \Lambda_{ij}^{(2)} = \frac{a_{32} - a_{23}}{a_{22} + a_{33}} \quad (15)$$

where the  $a$ 's are elements of the matrix  $\mathbf{A}_{ij}$  of eq. (4),  $\mu_i$  is the angle between the dipole-moment vector of peptide group  $p_i$  and the corresponding virtual-bond axis, and  $\mathbf{v}_i$  and  $\mathbf{v}_j$  are the unit vectors pointing from  $C_i^\alpha$  to  $C_{i+1}^\alpha$  and from  $C_j^\alpha$  to  $C_{j+1}^\alpha$ , respectively. The constant  $B$  in eqs. (6) and (7) depends on the type of peptide group (preceding proline or ordinary), and can be found in Table 2 of ref. 13.

The derivation of eq. (5) and of the expressions occurring in eqs. (8), (9), (10), (14), and (15) can be found in the Appendix of ref. 8 and in ref. 16.

As in our earlier work,<sup>13</sup> we assume a simple Lennard-Jones form for the interaction between peptide-group centers [eq. (16)]:

$$E_{\text{vdw};ij} = \epsilon_{ij} \left[ \left( \frac{r_{ij}^o}{r_{ij}} \right)^{12} - 2 \left( \frac{r_{ij}^o}{r_{ij}} \right)^6 \right] \quad (16)$$

The constants  $\epsilon_{ij}$  and  $r_{ij}^o$  depend on the type of the peptide group, and can be found in Table 2 of ref. 13. It should be noted that  $E_{\text{vdw}}$  is a constant for a fixed  $C^\alpha$  trace.

We express  $E_{\text{loc};i}$  as a Fourier series in  $\lambda^{(1)}$  and  $\lambda^{(2)}$  given by eq. (17). The coefficients depend on the type of residue; as in our earlier work,<sup>13,29</sup> we distinguish three types—glycine, proline, alanine—the latter denoting any residue except glycine or proline. The coefficients were calculated by fitting eq. (17) to the ECEPP/3 energy surfaces.

$$E_{\text{loc};k} = \sum_{i=0}^6 \sum_{j=0}^6 \{a_{k;ij} \cos[i \cdot \lambda_k^{(1)} + j \cdot \lambda_k^{(2)}] + b_{k;ij} \sin[i \cdot \lambda_k^{(1)} + j \cdot \lambda_k^{(2)}]\} \quad (17)$$

The bending energy is a simple harmonic potential, as expressed by eq. (18).

$$E_{\text{bend};i} = k(\tau_i - \tau_i^o)^2 \quad (18)$$

where  $k$  is a nonspecific force constant, which is proportional to the step number  $p$  (see Procedure section):

$$k = K_o \frac{p}{N} \quad (19)$$

[with  $K_o$  being an arbitrary constant equal to 100 in this work and  $N$  being the number of steps (see Procedure section)], and  $\tau^o$  is the value of the standard valence angle  $\text{N}-\text{C}^\alpha-\text{C}'$ , which depends on the type of amino-acid residue.<sup>30</sup>

### Procedure

For a given  $C^\alpha$ -trace the angles  $\lambda_1, \lambda_2, \dots, \lambda_{n-1}$  cannot be varied freely, because the constraints imposed by fixed values of the bond angles  $\tau_1, \tau_2, \dots, \tau_{n-1}$  must be obeyed. In this work, we, therefore, implement a procedure from our earlier article,<sup>8</sup> in which the  $C_i^\alpha-\text{C}'_i-\text{N}_{i+1}-\text{C}_{i+1}^\alpha$  atoms of the  $i$ th peptide group lie on a straight line and the  $O_i$  and  $H_{i+1}$  atoms are attached to  $C'_i$  and  $\text{N}_{i+1}$ , respectively, with right bond angles (Fig. 2b). This implies that the angles  $\xi$  and  $\eta$  defined in Figure 2a are set to zero and  $\tau_i = \theta_i$ ,  $i = 1, 2, \dots, n-1$  (Fig. 2b). In this case, the  $\lambda$ 's can vary freely without distorting the bond angles. We, therefore, start the optimization of peptide-group rotations by assuming a linear arrangement of the peptide-group atoms and, after optimization is accomplished with such linearity, we modify the  $\xi$ 's,  $\eta$ 's, and  $\tau$ 's as follows:

$$\xi_i^p = \frac{p}{N} \xi_i^o \quad (20)$$

$$\eta_i^p = \frac{p}{N} \eta_i^o \quad (21)$$

$$\tau_i^p = \frac{p}{N} \tau_i^o + \frac{N-p}{N} \theta_i \quad (22)$$

where  $N$  denotes the number of steps in transforming a linear peptide group into a peptide group with real geometry,  $p$  is the step number, and the superscript  $o$  denotes a standard ECEPP/3 value of the corresponding angle. In this work, we have arbitrarily assigned  $N = 10$ , which is a reasonable compromise between the smoothness of passing from one geometry of the (distorted) peptide group to another one and computational effort.<sup>8</sup>

At each of the consecutive steps  $p = 0, 1, \dots, N$ , optimization of the angles  $\lambda$  is carried out, with the energy function defined by eqs. (2)–(18) using the following Monte Carlo algorithm:

1. Initialize  $\lambda$  values. For  $p > 0$ , the  $\lambda$ 's of the preceding step are chosen as initial values; for  $p = 0$  a dipole-alignment algorithm is used, which is described later in this section (after step 7 of the procedure).
2. Choose a peptide group  $p_i$  for perturbation at random.
3. Save the value of  $\lambda_i$ .
4. Add random increment to the  $\lambda_i$ . The random increment is computed from eq. (23).

$$\Delta\lambda_i = (\text{rand} - 0.5) \times \Delta\lambda_{\max} \quad (23)$$

where rand is a random number from the interval  $[0, 1]$  and  $\Delta\lambda_{\max} = 0.42\pi$ ; this value has been determined by numerical experiments to achieve the fastest convergence of the MC procedure.

5. Calculate the difference between the energies of the unperturbed and perturbed configurations,  $\Delta E$ . It should be noted that, for  $p = 0$  (for linear peptide groups), the bending term is zero.
6. Accept the new value of  $\lambda_i$  if  $\Delta E < 0$ . Otherwise restore the old value of  $\lambda_i$ .
7. Iterate steps 2–6, until the energy does not decrease anymore, or a maximum of 10,000 steps have been tried.

Clearly, the positions of the  $C^\alpha$  atoms remain unchanged throughout the procedure.

To determine a reasonable initial configuration of dipoles, we use the concept of our dipole path method<sup>8</sup> and try to construct as many paths of aligned dipoles as possible. It should be noted, however, that while the dipole path network is absolutely essential for the dipole path method, here it serves only to determine a reasonable starting configuration of the dipoles, and the bulk of the overly complicated structure of the algorithm for the construction of an *accurate* dipole path network is avoided. To determine the dipole path network, we take advantage of the well-known triangle inequality. We examine all triplets of peptide-group centers ( $p_i, p_j, p_k$ ), located farther than one residue away in the sequence. If both distances  $d_1 = d(p_i, p_j)$  and  $d_2 = d(p_j, p_k)$  are lower than 4.5 Å, which we consider as a boundary for the distances between hydrogen-bonded peptide groups, and the difference between the sum of these distances and the distance between  $p_i$  and  $p_k$ ,  $\Delta d = |d(p_i, p_j) + d(p_j, p_k) - d(p_i, p_k)|$  is lower than 0.4 Å (if this holds,  $p_i, p_j$ , and  $p_k$  are, within a good approximation, almost on

a straight line), we accept the  $p_i, p_j, p_k$  sequence. Then, we merge all sequences of dipoles that share peptide groups to form dipole paths. Subsequently, we align peptide-group dipoles along the paths, using the alignment procedure of the dipole-path method.<sup>8</sup> As shown in our previous work,<sup>8</sup> there are two directions along which the dipoles in a path can be aligned; we choose the direction with lower local-interaction energy. This procedure is faster than calculation of the full energy and, in most cases, produces reasonable starting configurations of dipoles.

For those peptide groups that cannot be assigned to dipole paths and in cases in which the assignment to dipole paths is ambiguous, we apply the following formula:

$$\lambda_i = -0.45\pi \quad (24)$$

which has been determined by numerical experiments. For all but the last full residue  $\lambda_i = \lambda_i^{(1)}$ , which implies that the local conformational state of the corresponding amino acid residue is chosen from a low-energy region. (A negative value of  $\lambda^{(1)}$  implies a negative value of the dihedral angle  $\phi$ , which implies a low-energy region of the  $\phi, \psi, \text{map}^{31}$ ).

As in our earlier work,<sup>8</sup> proline residues are treated separately, because of the constraint imposed by the presence of the pyrrolidine ring; they are calculated to achieve  $\phi_{\text{pro}} = -68.3^\circ$ , which corresponds to the ECEPP/3 value.<sup>27</sup>

After optimization of the simplified energy function [eq. (2)] is completed using the above-described procedure, the structure is subjected to constrained energy minimization with the ECEPP/3 force field.<sup>27</sup> The constraints are all distances between  $C^\alpha$  atoms. The penalty function is expressed by eq. (25).

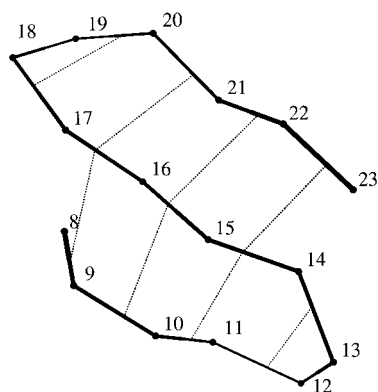
$$f_{\text{pen}} = w_{\text{pen}} \frac{1}{4} \sum_{i=1}^n \sum_{j=1}^{n-1} \begin{cases} (d_{ij}^2 - d_{ij}^o)^2 & \text{if } |d_{ij} - d_{ij}^o| > 1 \text{ Å} \\ 0 & \text{otherwise} \end{cases} \quad (25)$$

where  $d_{ij}$  is the distance between  $C_i^\alpha$  and  $C_j^\alpha$  and  $d_{ij}^o$  is the corresponding distance in the original  $C^\alpha$  trace. Three consecutive minimizations are carried out with  $w_{\text{pen}} = 1.0, 0.1$ , and 0.01, respectively. To save computational time, we substitute alanine for all amino acid residues except glycine and proline.

## Results and Discussion

In the following subsections we present three test cases: Ac-Ala<sub>10</sub>-NHMe in  $\alpha$ -helical conformation, the 16-residue fragment of betanova,<sup>32</sup> a designed 20-residue  $\beta$ -sheet peptide, whose structure has been determined by NMR (PDB code: 1BH35), and the 61-residue immunoglobulin binding protein (an  $\alpha/\beta$ -protein; PDB code: 1IGD<sup>33</sup>). The  $C^\alpha$  trace of the first one was obtained from the all- $\alpha$ -helical structure subjected to energy minimization with the ECEPP/3 force field, while the  $C^\alpha$  traces of the second and the third ones were taken from the PDB coordinates without optimization.





**Figure 4.** The dipole path network (dotted lines) superposed on the  $C^\alpha$  trace of betanova. Residue numbering according to the NMR structure.<sup>32</sup>

#### *Ac-Ala<sub>19</sub>-NHMe (full $\alpha$ -helix)*

There are three dipole paths detected, based on hydrogen-bonding contacts between peptide groups, namely 20-17-14-11-8-5-2, 19-16-13-10-7-4, and 18-15-12-9-6-3. The lowest local-interaction energy is achieved when the dipoles point along sequences of the paths, as in  $\alpha$ -helices. The Monte Carlo search for optimal dipole alignment described in the Methods produces average values of the dihedral angles  $\phi$  and  $\psi$  equal to  $-57.0^\circ$  and  $-49.9^\circ$ , respectively; these values are already in the  $\alpha$ -helical A region of the conformational energy map of Zimmerman et al.<sup>31</sup> After energy minimization with the ECEPP/3 force field, the dihedral angles  $\phi$  and  $\psi$  become  $-67.5^\circ$  and  $-39.0^\circ$ , respectively, which are values characteristic of polyaniline  $\alpha$ -helices minimized with the ECEPP/3 force field.

#### *Betanova (Three-Stranded Antiparallel $\beta$ -Sheet)*

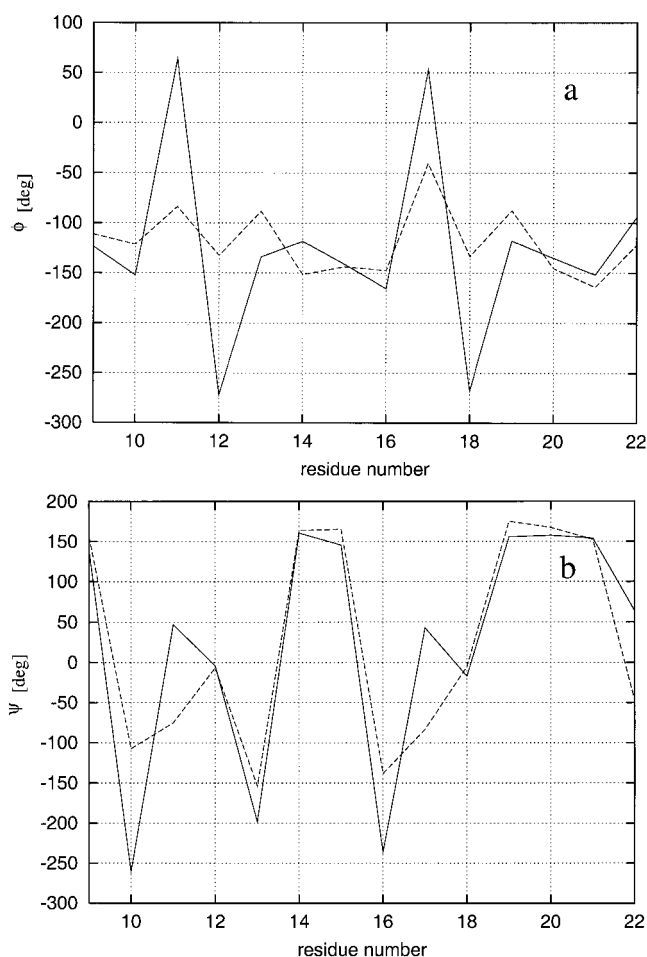
The one-letter code of the part of the sequence of the peptide present in the NMR structure<sup>32</sup> is WSVQNGKYTNNGKTTE; the residue numbering in the NMR structure starts with 8. The structure is a three-stranded antiparallel  $\beta$ -sheet with turns at Asn<sup>12</sup>–Gly<sup>13</sup> and Asn<sup>18</sup>–Gly<sup>20</sup>.<sup>32</sup> The dipole paths based on hydrogen-bonding contacts are 9-17-21, 22-16-10, 11-15-23, 20-18, and 14-12. The lowest local-interaction energy was obtained for the orientation of dipoles along the enumerated path, which is characteristic of an antiparallel  $\beta$ -sheet; in particular, the dipoles at  $\beta$ -turns were directed from peptide group 14 to peptide group 12 and from peptide group 20 to peptide group 18, respectively. The paths are shown in Figure 4.

The dihedral angles  $\phi$  and  $\psi$  resulting from our procedure are compared with those calculated from the NMR structure in Figure 5, and the predicted backbone is superposed on its counterpart from the NMR structure in Figure 6. The agreement between the predicted and experimental dihedral angles  $\phi$  and  $\psi$  is good, and the structures superpose very well except for some differences in the turn regions. The correlation coefficients between the predicted and experimental dihedral angles  $\phi$  and  $\psi$  are 0.65 and 0.87, respectively. The discrepancies in turn regions are caused by the

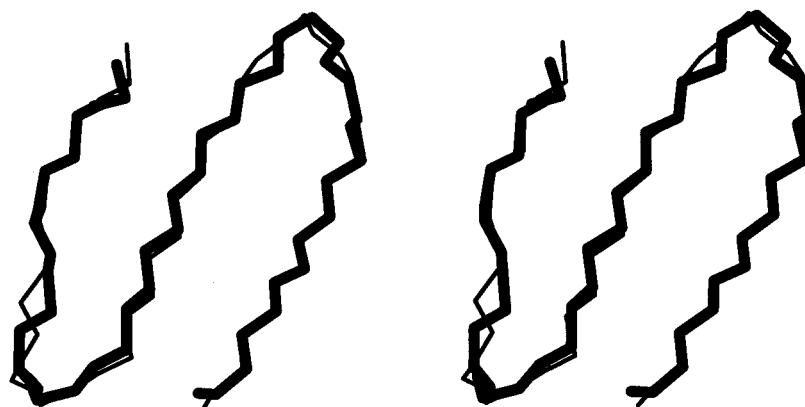
fact that the NMR structure has type I'  $\beta$ -turns at residues Asn<sup>12</sup>–Gly<sup>13</sup> and Asn<sup>18</sup>–Gly<sup>20</sup>, with Asn<sup>12</sup> and Asn<sup>18</sup>, respectively, lying in the unfavorable A\* region<sup>31</sup> of the energy map. The presence of an energetically unfavorable type I'  $\beta$ -turn might be an artifact of the force field used in the refinement of the NMR structure. The predicted backbone contains type I  $\beta$ -turns, with Asn<sup>12</sup> and Asn<sup>18</sup> lying in the energetically favorable A region<sup>31</sup> of the energy map.

#### *Immunoglobulin-Binding Protein (an $\alpha/\beta$ Protein)*

The structure of IIGD contains a four-stranded  $\beta$ -sheet (formed from the N- and the C-terminal parts of the protein) packed against an  $\alpha$ -helix formed by residues Ala<sup>28</sup>–Asp<sup>41</sup>. The recognized sequences of dipoles forming the dipole path network are as follows: 26-6, 7-25, 24-8, 52-56-9-23, 22-10-57-51, 50-58-11-21, 20-12-59-49, 19-13-60-48, 18-14-61, 55-53 (forming the dipole paths of the  $\beta$ -sheet part), and 41-38-35-32-29, 42-39-36-33-30, 40-37-



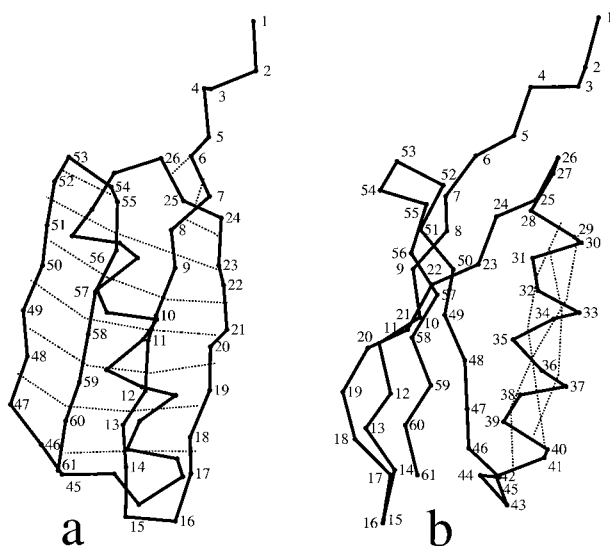
**Figure 5.** Comparison of the dihedral angles  $\phi$  (a) and  $\psi$  (b) of betanova calculated from the NMR structure<sup>32</sup> (solid lines) and predicted by our procedure (dashed lines). The correlation coefficients between the experimental and predicted dihedral angles are 0.65 for  $\phi$  and 0.87 for  $\psi$ , respectively. Residue numbering follows that of the NMR structure.



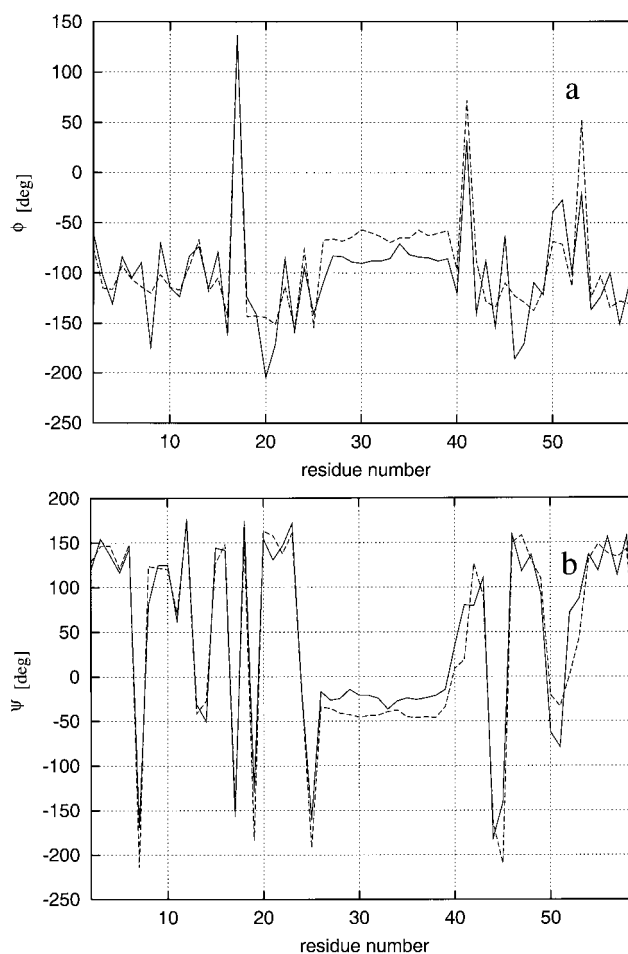
**Figure 6.** Stereoview of the superposition of the backbone of the NMR structure of betanova<sup>32</sup> (thick line) on the structure resulting from the application of our procedure (thin line). The rmsd over backbone atoms is 0.54 Å.

34-31 (forming the dipole paths of the  $\alpha$ -helix). These paths are shown in Figure 7.

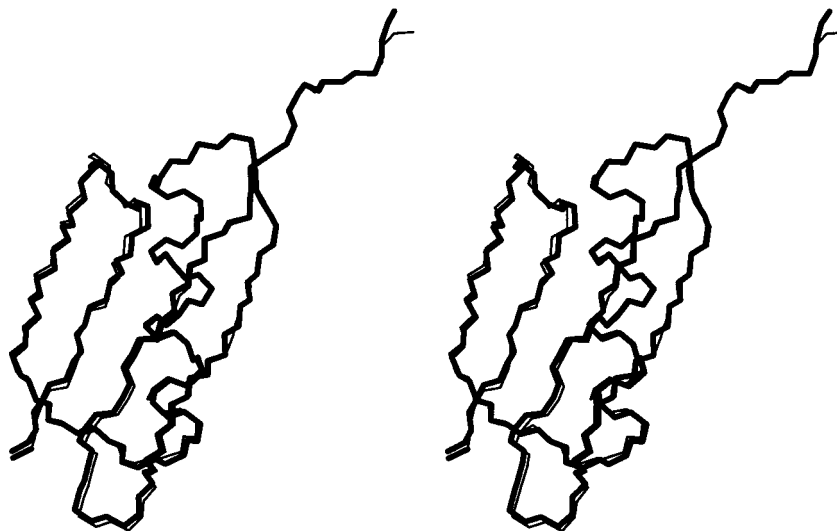
The dihedral angles  $\phi$  and  $\psi$  resulting from the alignment procedure and from the subsequent constrained energy minimization are compared with the values calculated from the crystal structure in Figure 8, while the superposition of the backbone from the crystal structure and the backbone resulting from the reconstruction is presented in Figure 9. The rms deviation over all backbone atoms is 0.50 Å, the maximum deviation being 0.7 Å for Met<sup>1</sup>, Thr<sup>21</sup>, Thr<sup>58</sup>, and Lys<sup>59</sup>. Such rmsd values are typically obtained when fitting crystal structures to rigid-valence ECEPP/3 geometry.<sup>8</sup> It should be noted that the predicted and experimental dihedral angles  $\phi$  and  $\psi$  are in very good correlation with those



**Figure 7.** The dipole path network of IIGD (dotted lines) superposed on the C $\alpha$  trace of its crystal structure.<sup>33</sup> (a) the part corresponding to the  $\beta$ -sheet; (b) the part corresponding to the  $\alpha$ -helix.



**Figure 8.** Comparison of the dihedral angles  $\phi$  (a) and  $\psi$  (b) of IIGD calculated from the crystal structure<sup>33</sup> (solid lines) and predicted by our procedure (dashed lines). The correlation coefficients between the experimental and predicted dihedral angles are 0.97 for  $\phi$  and 0.97 for  $\psi$ .



**Figure 9.** Stereoview of the superposition of the backbone of the crystal structure of 1IGD<sup>33</sup> (thick line) on the structure resulting from the application of our procedure (thin line). The rmsd over backbone atoms is 0.50 Å.

calculated from the crystal structure, the correlation coefficient being 0.86 and 0.97, respectively.

#### Timing

The computation time ranged from 3 min for the Ac-Ala<sub>19</sub>-NHMe system to 22 min for 1IGD; for proteins with about 150 amino acid residues that we treated in the recent CASP4 exercise,<sup>10</sup> the computation time was about 2 h. All these data refer to a single processor of a 650 MHz Pentium III running Linux.

#### Conclusions

The algorithm for reconstruction of a polypeptide backbone from its C $\alpha$  trace, described in this work, was shown to be capable of restoring the correct backbone geometry within a 0.4–0.6 Å rmsd, which is of comparable quality to that of fitting crystal structures to rigid-valence geometry or refinement of crystal structures by molecular dynamics. Moreover, the predicted dihedral angles  $\phi$  and  $\psi$ , which define the local conformational states, are in excellent correlation with the values calculated from the original experimental structures. The method is based fully on energy optimization, with no reference made to structural data bases. The method implements a simplified energy function in the first stage to facilitate the search for optimal rotations of the peptide groups to finish the refinement using local minimization of an all-atom energy function. Because of explicit inclusion of local interactions in the simplified energy function, it handles loop regions and other regions with nonregular structures well. In this regard, it is superior to our earlier designed dipole path method,<sup>8</sup> which optimized only electrostatic interactions and performed significantly worse in loop regions; moreover, because local interactions were neglected, sophisticated rules for dipole path construction had to be designed.<sup>8</sup>

Apart from the test cases described, we have already applied the method in the last CASP4 exercise<sup>10</sup> to convert coarse-grain structures predicted with the UNRES force field into all-atom structures. We are currently working on extending the method to reconstruct the complete all-atom geometry from its C $\alpha$  trace and the positions of side-chain centers, which information is contained in coarse-grain structures predicted with the UNRES force field. Another issue is the need to reduce computational time, which is remarkably larger for larger chains. There are two possible ways to reduce computation time: (i) by applying a distance cutoff in the evaluation of the dipole-interaction energy (because the positions of the  $\alpha$ -carbon atoms are fixed, the cutoff list will be fixed); and (ii) by applying the Self-Consistent Electric Field (SCEF) procedure in sampling peptide-group rotations.<sup>22</sup> These issues are now being addressed in our laboratory.

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